FT Wilshire GLIO Listed Infrastructure Index Series: Capping infrastructure indexes

By Andrew Dougan

The FT Wilshire GLIO Listed Infrastructure Index Series provides exposure to globally listed companies which own and/or operate infrastructure assets. Wilshire consults with the Global Listed Infrastructure Organization (GLIO), which represents the $3tn listed infrastructure asset class.

Executive Summary

With growth in infrastructure speeding up globally, the importance investors place on sectors, countries, and regional exposure is emerging as a key theme in infrastructure allocations. The growing weight of US equities and regulated utilities in standard infrastructure indexes has led many global investors to seek an infrastructure index series that offers more diversification.

To this end, the FT Wilshire GLIO Listed Infrastructure Index series has been expanded to include three new capped indexes based on the developed markets' portion of the broader global index. We examine the properties of these capped indexes and compare them to their uncapped counterparts.

At the highest level, infrastructure companies are those in the categories shown in Figure 1. Social Infrastructure, such as hospitals, care homes and schools, or companies better classified as real estate operating companies, are excluded.

Wilshire Indexes uses its Global Asset Taxonomy System (GATS) to identify the companies in these sectors. This is achieved by mapping GATS Level 4 to the infrastructure sectors in Figure 1. However, not all companies will be relevant, so additional means of screening are necessary. To further disqualify companies, Wilshire Indexes uses Natural Language Processing (NLP) on company accounts data, EBITDA ratios, asset values, and sales revenues. The FT Wilshire Global Equity Markets Series (GEMS) provides a suitable universe to base the global infrastructure series. GEMS is a comprehensive

measure of the global stock markets, covering up to 98% of investable market capitalization in 48 developed and emerging countries. GEMS is built country-by-country while balancing the size integrity among developed and emerging markets through a consistent and modular-based methodology. This facilitates a variety of carve-outs for the infrastructure series, e.g. by size, country, or region.

Headline index methodology

The FT Wilshire GLIO Listed Infrastructure Index Series uses a four-step systematic approach to determine company selection and eligibility:

1. Natural Language Processing (NLP) techniques are used to identify candidate companies and isolate infrastructure-related EBITDA
   a. Applied to each infrastructure sector defined by GATS[1]
   b. Companies and EBITDA are identified via an accept/reject of relevant words and phrases

2. Candidate companies must have infrastructure-related EBITDA ratio of 0.66 or more to be eligible
   c. If EBITDA is unknown, asset values are considered
   d. If there is no asset value breakdown, sales revenue is used
   e. If sales revenue is unavailable, the Wilshire-GLIO advisory group can provide oversight

3. Apply 10% EBITDA & liquidity buffer on existing constituents to prevent unnecessary turnover and potential whipsawing of index membership.

Wilshire-GLIO advisory group provides additional oversight of index membership[2]. Once the list of companies is obtained, constituents are weighted by free-float market capitalization in US dollars. The indexes are reviewed semi-annually in March and September.

Variants where countries, sectors and stocks can be capped at a maximum weight are discussed in the next section. Further customization options include, equal weighting of infrastructure sectors and exclusion/inclusion indexes, e.g. excluding US or else including data centers[3].

Figure 1: Infrastructure Sectors
Source: FT Wilshire

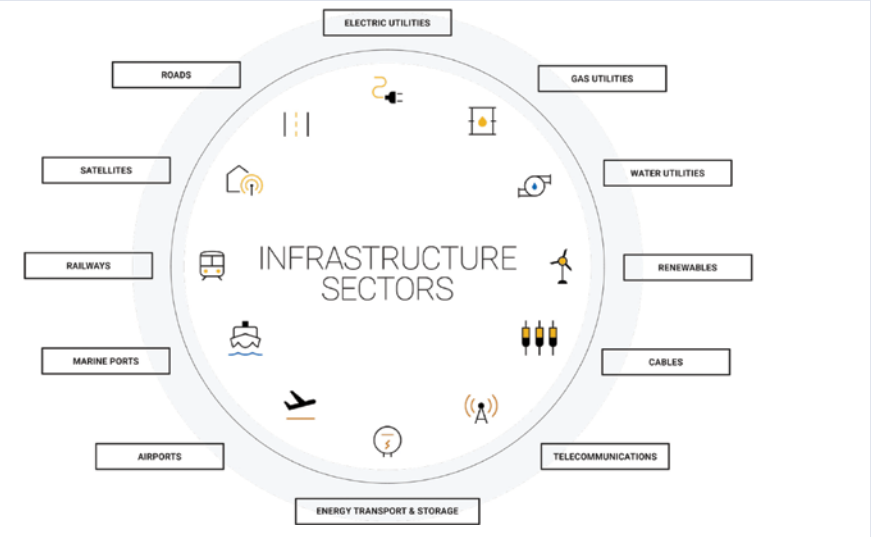
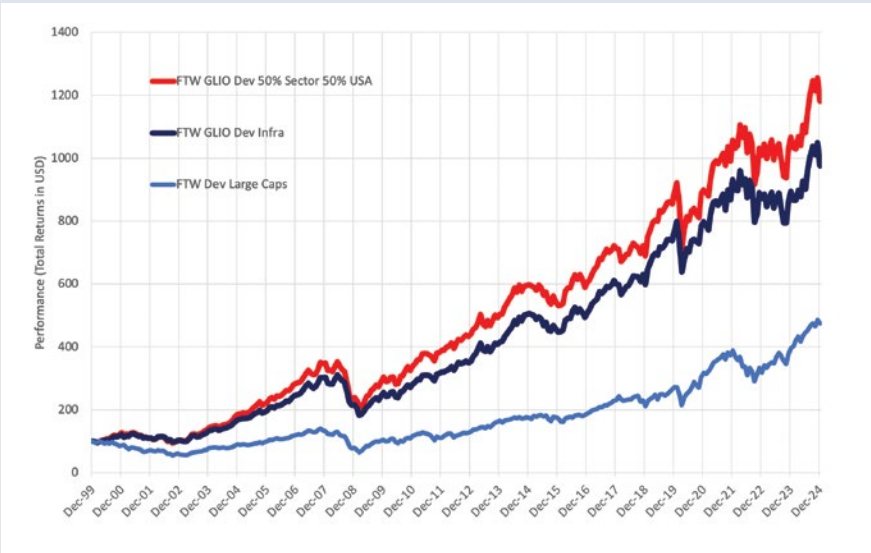


Figure 2: Index Performance: FT Wilshire Developed Index, FT Wilshire GLIO Developed Listed Infrastructure Index and FT Wilshire GLIO Developed Listed Infrastructure Capped 50% Sector & USA Index



Capped infrastructure indexes

To illustrate the series, we concentrate on two of our developed products; the FT Wilshire GLIO Developed Listed Infrastructure Index and a capped version, the FT Wilshire GLIO Developed Listed Infrastructure Capped 50% Sector & USA Index.

This latter index is designed for global investors who wish to cap the weight of the US at 50%, stocks at 5%, and sectors such that:

1. Utilities (inc. Renewables) capped to 50% by weight >

1 The GATS sectors are mapped over to FT Wilshire GLIO infrastructure sectors. The mapping scheme is available here: https://cdn.prod.website-files.com/63e3e50fdce0bcaff7861530/66a38b0a1d678a43e5ca11e6_FT%20Wilshire%20GLIO%20Listed%20Infrastructure%20Index%20Series%20Methodology.pdf

2 The Wilshire-GLIO advisory group meets regularly to discuss issues, and developments on the index series. Regular contact with the group ensures the index is representative for current market conditions and future developments.

3 The inclusion of Data Centers has been a recent discussion topic with the Wilshire-GLIO advisory group.

Table 1: Annualized Performance Statistics - FTW Dev Index, FTW GLIO Dev Infra Index and FTW GLIO Dev Infra Capped 50% Sector & USA Index

Source: FTW GLIO, FT Wilshire

USD Total Returns (Yrs Annualized)	2024	2023	3 Years	5 Years	7.5 Years	10 Years	12.5 Years	15 Years	17.5 Years	20 Years	22.5 Years
FTW GLIO Dev Infra	9.2%	2.5%	1.6%	4.9%	7.5%	6.8%	8.8%	9.4%	7.5%	9.2%	10.1%
FTW GLIO Dev 50% Sector 50% USA	10.7%	5.9%	3.8%	5.8%	7.7%	7.1%	8.8%	9.5%	7.8%	9.7%	11.2%
FTW Developed Large Caps	19.4%	24.4%	6.8%	11.7%	11.6%	10.6%	11.7%	10.6%	7.6%	8.7%	9.1%

Investors can use listed infrastructure to build a separate dedicated exposure to the asset class without significant overlaps with their broad equities allocation.

2. Transportation capped at 20%
3. Railways capped at 6.67% (1/3 Transportation weight)
4. Energy capped to 20%
5. Communications capped to 10%

Figure 2. displays the performance of the infrastructure indexes between March 1999 and December 2024 against their cap weighted benchmark, the FT Wilshire Developed Index.

Both infrastructure indexes have outperformed their market cap-weighted benchmark over the long term. However, in the last ten years, the market capitalization index has been dominant. This can be seen in the performance statistics in Table 1.

No doubt, this behavior has been aided by the strong performance of the tech sector, which of course has zero weight in the infrastructure indexes.

Table 2. shows that the FT Wilshire GLIO Developed Listed Infrastructure Index is already underweight US compared to the market capitalization index, but the capped index goes further and constrains this to be no more than 50% of the global weight at rebalance.

Table 3. emphasizes the 50% constraint on the weight of Utilities and the 20% limit on Transportation in the capped index.

Table 2: Country Weights: FTW Dev Index, FTW GLIO Dev Infra Index and FTW GLIO Dev Infra Capped 50% Sector & USA Index

Source: FT Wilshire, GLIO

Country	FTW Developed. Index	FTW GLIO Dev. Listed Infra. Index	FTW GLIO Dev. Listed Infra. Capped 50% Sector & USA Index
Australia	**1.83%**	**1.84%**	**5.78%**
Austria	0.08%	0.22%	0.23%
Belgium	0.21%	0.08%	0.08%
Canada	3.18%	13.33%	15.59%
Switzerland	2.54%	0.16%	0.76%
Germany	2.00%	1.80%	2.24%
Denmark	0.70%	0.32%	0.34%
Spain	0.66%	5.44%	10.89%
Finland	0.25%	0.00%	0.00%
France	2.33%	1.45%	1.98%
United Kingdom	3.69%	3.75%	4.04%
Hong Kong	0.51%	1.47%	1.59%
Ireland	0.28%	0.00%	0.00%
Israel	0.26%	0.07%	0.08%
Italy	0.67%	2.95%	3.30%
Japan	5.83%	0.75%	0.40%
Luxembourg	0.18%	0.05%	0.09%
Netherlands	1.16%	0.11%	0.19%
Norway	0.17%	0.00%	0.00%
New Zealand	0.08%	0.22%	0.85%
Portugal	0.04%	0.40%	0.43%
Singapore	0.46%	0.05%	0.09%
Sweden	0.83%	0.00%	0.00%
United States	72.05%	65.57%	51.05%
Total	100%	100%	100%

Table 3: Sector Weights - FTW GLIO Dev Infra Index and FTW GLIO Dev Infra Capped 50% Sector & USA Index

Sector Group Weights	FTW GLIO Dev Infra	FTW GLIO Developed 50% Sector & USA Capped
Communications	6.28%	8.45%
Energy Transportation	16.55%	22.92%
Transportation	18.30%	19.73%
Utilities & Renewables	58.86%	48.91%
Total	100.00%	100.00%

Table 4: Infrastructure Sector Weights – FT Wilshire GLIO Developed Listed Infrastructure Index and FT Wilshire GLIO Developed Listed Infrastructure Capped 50% Sector & USA Index
Source: FT Wilshire, GLIO

Sector Weights	FT Wilshire Global Dev Equities	FTW GLIO Dev Infra	FTW GLIO Developed 50% Sector & USA Capped
Electric Utilities	1.47%	42.91%	34.14%
Energy T&S	0.49%	16.55%	22.92%
Rail	0.57%	15.37%	6.14%
Renewables	0.29%	7.37%	7.90%
Gas Utilities	0.21%	6.31%	4.98%
Telecom Infra	0.23%	6.00%	7.96%
Water Utilities	0.08%	2.27%	1.89%
Roads	0.07%	1.83%	8.58%
Airports	0.04%	1.10%	5.01%
Satellites	0.01%	0.17%	0.26%
Cables	0.00%	0.11%	0.22%
Total	3.46%	100.00%	100.00%
Non Infrastructure	96.54%		

The upshot of this multi-level capping is shown in Table 4 – the FT Wilshire GLIO Developed Listed Infrastructure Capped 50% Sector & USA Index has a more balanced set of infrastructure sector weightings than the FT Wilshire GLIO Developed Listed Infrastructure Index.

Note the lower weight in Railways for the capped index. The table shows very little overlap (3.5%) between broad equities and the listed infrastructure carve-out. This means that investors can use listed infrastructure to build a separate dedicated exposure to the asset class without significant overlaps with their broad equities allocation.

The effective number of infrastructure sectors, given by the Herfindahl measure (sum of inverse square weights), is approximately four for the uncapped index and five for the capped index. Hence, improved diversification across sectors is delivered.

Summary

The FT Wilshire GLIO Listed Infrastructure Index Series provides exposure to globally listed companies which own and/or operate infrastructure assets. The indexes use a systematic approach to identify eligible infrastructure companies using EBITDA to determine infrastructure relevance and by mapping the Global Asset Taxonomy System (GATS) to Infrastructure Sectors. The series allows customization through size, country and regional carve-outs of the GEMS universe.

The series contains three capped variants:

1. FT Wilshire GLIO Developed Listed Infrastructure Capped 50% USA Index
2. FT Wilshire GLIO Developed Listed Infrastructure Capped 50% Sector Index
3. FT Wilshire GLIO Developed Listed Infrastructure Capped 50% Sector & USA Index

These are suitable for investors seeking extra diversification by either dampening the growing market cap weight of US equites, reducing the weight of the regulated utility sector group or by imposing both limits simultaneously.

The indexes use a systematic approach to identify eligible infrastructure companies using EBITDA to determine infrastructure relevance and by mapping the Global Asset Taxonomy System to Infrastructure Sectors.

Want more information?
Join Wilshire Indexes' CEO, Mark Makepeace, at the GLIO/ Morgan Stanley Sydney seminar on 18 March, 2025. Register here: www.glio.org/event-details/sydney25

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Andrew Dougan is a Managing Director at Wilshire Indexes with responsibility for index research. He has a background in factor analysis and model creation, portfolio optimization, attribution analysis, equity derivatives and is currently involved in the design and implementation of various types of infrastructure, smart beta, factor, and climate indexes.

At the core of a multi-asset portfolio – listed infrastructure

By Chuck DIROCCO & Fraser HUGHES

This is an update to the original article we penned in June 2022. Since then, global equities have outpaced global infrastructure over the last two full calendar years. This presents an opportune moment to examine how recent performance trends have altered the composition of a multi-asset portfolio.

This study aims to provide a comprehensive overview, demonstrating that a portfolio of listed infrastructure companies serves as an effective diversifier. It highlights how such a portfolio can increase annual returns and lower overall risk within a multi-asset investment strategy.

Using a simple modern portfolio theory framework for our analysis, time series included:

- US T-Bills (risk-free rate)
  - short-term debt securities, 3-month[CD1] [FH2] , issued by the US Department of the Treasury
- Global Bonds – Bloomberg Global Aggregate USD[1]
  - a comprehensive measure of the global investment-grade fixed-income market
- FT Wilshire Developed Large Caps – USD[2]
  - A global index tracking the performance of large-cap, developed market stocks.
- FT Wilshire GLIO Developed Listed Infrastructure Index[3]
  - designed to measure the performance of globally listed 'pure-play' infrastructure companies.

1 www.bloomberg.com/quote/LEGATRUU:IND

2 In this update, we replace MSCI World, with FT Wilshire Developed Large Caps. The performance of the indexes is identical over the short, medium, and long term.

3 In this update, we replace the legacy GLIO Index with the FT Wilshire GLIO Developed Listed Infrastructure Index.

Listed infrastructure offers downside resilience, stable yields, and reduced correlations to traditional asset classes.

Overall results make a strong case for listed infrastructure to be included in any diversified investment portfolio. Listed infrastructure offers several compelling benefits, including downside resilience, stable yields, and reduced correlations to traditional asset classes. We believe that by incorporating listed infrastructure, investors can achieve a more balanced and robust portfolio, capable of weathering market volatility while delivering consistent returns.

The analysis

Firstly, we ran 10,000 randomly weighted portfolios of global bonds, global equities, FTW GLIO listed infrastructure companies, and US T-Bills over a long-term 20-year investment horizon: December 2004 to December 2024. The risk/return profile of these portfolios is indicated by the red dots in Figure 1.

Within this array of portfolios, we've highlighted (in red) the four asset class risk/return profiles and an efficient frontier (the red line in the center of the chart). This efficient frontier comprises 40% global bonds, 10% US T-Bills, and a 50% allocation to global equities and listed infrastructure. Figure 2. focuses on the efficient frontier shown in Figure 1. Moving from right to left, we see portfolio risk decrease by adding 5% allocations to listed infrastructure at the expense of global equities. Figure 3. illustrates the same impact. >

Figure 1: 20-year total returns and risk of 10,000 multi-asset portfolios (2004 to December 2024)

Source: FT Wilshire, FTW GLIO, Bloomberg, US T-Bills: https://fred.stlouisfed.org/series/DTB3

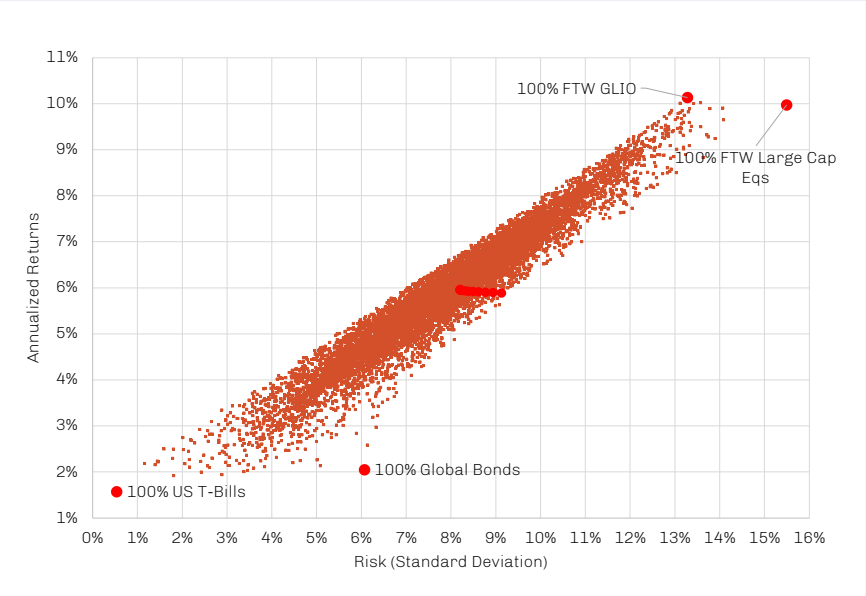
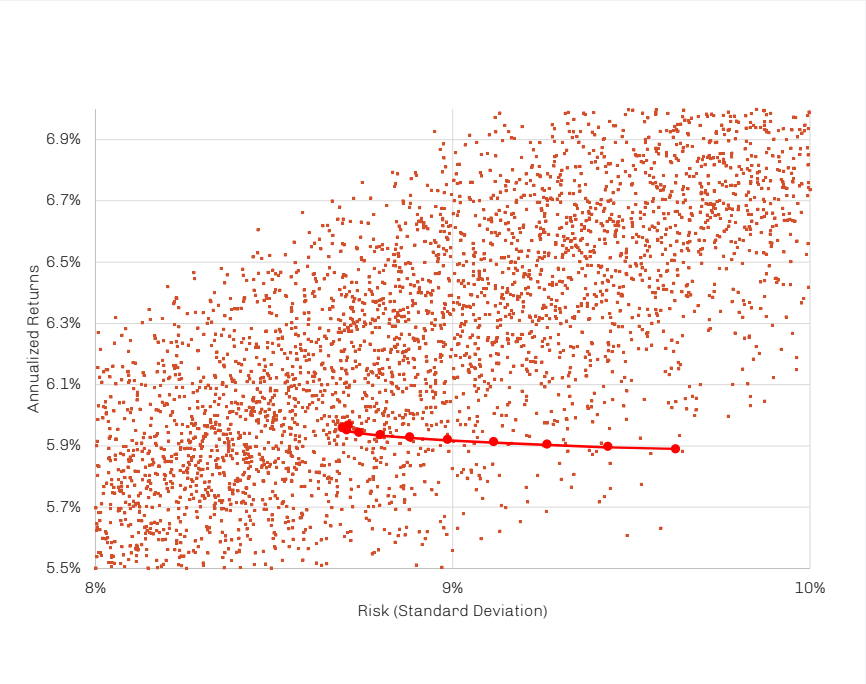


Figure 2: Magnified view of figure 1, focused on the efficient frontier created by a 50% allocation to global equities and listed infrastructure

Source: FT Wilshire, FTW GLIO, Bloomberg, US T-Bills



Expanding the example, Figure 4. displays seven possible portfolios ranging from highest return (100% FTW GLIO) to lowest risk (100% US T-Bills), with various portfolio mixes in between these extremes. Similar to our initial findings, the results display that increasing allocations to listed infrastructure from zero (Portfolio 3), 15% (Portfolio 6), 25% (Portfolio 5), 45% (Portfolio 7), and up to 50% (Portfolio 4) lowers portfolio risk and increases annual returns. The modified Sharpe ratio is highlighted in the column to the right.

The higher modified Sharpes indicate that an allocation to listed infrastructure, at the expense of global equities, enhances portfolio performance. While a 50% allocation to listed infrastructure is highly unlikely in the real world,[4] it is clear from these results that an allocation to listed infrastructure has benefits at any level in a multi-asset portfolio.

Finally, running simple, efficient frontiers that blend global equities and listed infrastructure over multiple investment time horizons shows that listed infrastructure is always included in the mix. The 20-year results align with the broader analysis, while the 5-, 10- and 15-year also demonstrate that listed infrastructure exposure benefits the broader portfolio.

The advantages

In previous issues of the *GLIO Journal*, we've consistently shown that allocations in listed infrastructure and unlisted (or private) infrastructure are complementary. After all, we are essentially looking at a similar pool of underlying assets, subject to the same regulation and economic conditions. Consequently, the long-term performance of the two forms of ownership is naturally very similar (see Figure 6).

4 *In NASRA's latest US public fund survey (Nov 24), the average public pension fund is as follows: public equities 44%, fixed income 21%, real estate 8%, alternative investments (including private equity, hedge funds and infrastructure) 26%, and cash 2%.*

The modified Sharpe ratios reveal that a 50% allocation to listed infrastructure, replacing global equities, optimizes the risk-adjusted return of a multi-asset portfolio over a 20-year horizon.

Figure 3: How increasing allocations to listed infrastructure enhance the multi-asset modified Sharpe ratio

Source: FT Wilshire, FTW GLIO, Bloomberg, US T-Bills

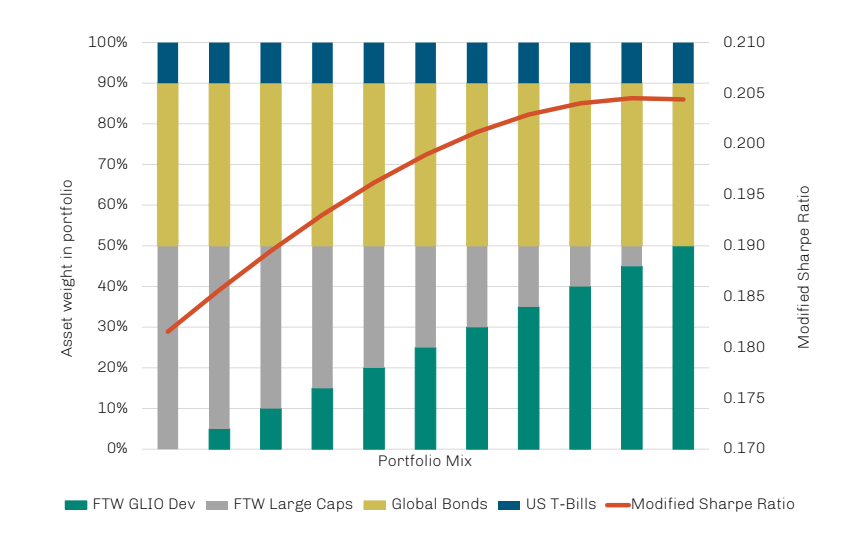


Figure 4: Portfolio mixes – maximum return to minimum risk (2004 to December 2024)

Source: FT Wilshire, FTW GLIO, Bloomberg, US T-Bills

Portfolio	Global Bonds	US T-Bills	FTW Equities	FTW GLIO	Risk	Return	Modified Sharpe
1 – Highest return	0	0	0	100	13.2%	10.2%	0.21
2 – Lowest risk	0	100	0	0	0.5%	1.6%	0.84
3 – 50% Equities	40	10	50	0	9.1%	5.9%	0.18
4 - 50% FTW GLIO	40	10	0	50	8.2%	6.0%	0.20
5 – Equal Weighted	25	25	25	25	7.8%	5.9%	0.21
6 – 15% FTW GLIO	35	10	40	15	8.6%	5.4%	0.18
7 – Efficient Fronter (45% FTW GLIO)	40	10	5	45	8.1%	6.0%	0.20